

Quantitative Aspects of Crop Production

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Chapter 6(A): Lintul-1: potential crop growth

A simple general crop growth model
for optimal growing conditions
(example: spring wheat)

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Contents:

LINTUL1 A simple general crop growth model for optimal growing conditions

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Introduction	6A -1
The main model equations and functions of the program	6A -1
Light interception	6A -1
Radiation or Light use efficiency	6A -2
Crop development	6A -3
Allocation of assimilates	6A -4
Leaf area growth.....	6A -5
Leaf area senescence	6A -6
Start conditions and environmental conditions.....	6A -7
Explanation of the simulation program	6A -7
1. Initial conditions and run control.....	6A -8
2. Environmental data and temperature sum	6A -8
3. Leaf growth and senescence	6A -9
4. Light interception and total crop growth rate	6A -10
5. Growth rates and dry matter production of plant organs.....	6A -11
6. Functions and parameters for spring wheat.....	6A -11
Reference to literature 'Introduction to FST'	6A -13
Exercises with LINTUL1.	6A -14
Table with DAYNUMBER OF YEAR (DOY):	6A -20
Handouts: Lintul1 - Potential crop growth	6A -21

LINTUL1 A SIMPLE GENERAL CROP GROWTH MODEL FOR OPTIMAL GROWING CONDITIONS (EXAMPLE: SPRING WHEAT)

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Introduction

LINTUL1 simulates potential growth of a crop, i.e. its dry matter accumulation under ample supply of water and nutrients in a pest-, disease- and weed-free environment, under the prevailing weather conditions. The rate of dry matter accumulation is a function of irradiation and crop characteristics. The model makes use of the common observation that the crop growth rate under favourable conditions is proportional to the amount of light intercepted (Monteith, 1977). Dry matter production is, therefore, modelled as the product of light interception and a constant light use efficiency. The dry matter produced is partitioned among the various plant organs, using partitioning factors defined as a function of the phenological development stage of the crop. The dry weights of the plant organs are obtained by integration of their growth rates over time.

LINTUL1 requires as input physiological properties of the crop (in this case for spring wheat) and the actual weather conditions at the site, characterized by its geographical latitude, i.e. daily maximum and minimum temperatures and irradiation for each day of the year.

The main model equations and functions of the program

LINTUL1 is based on a few mathematical equations and functions that describe the processes leading to dry matter production and grain yield. These equations and functions are described in this section. They can be used in a computer program that automatically computes crop dynamics throughout the season. For those who wish to understand the program implementation, the simulation program is fully explained after the description of the main model equations.

Light interception Radiation, more particularly the photosynthetically active radiation, provides the energy for the process of photosynthesis. The interception of radiation by the crop is thus essential. Assuming a homogeneous canopy, light exponentially decreases with the cumulative leaf area index, counted from the top of the canopy to the soil surface according to Lambert-Beer's law:

$$I = I_0 e^{-kL} \quad (1)$$

Where, I_0 is the incident radiation flux of photosynthetically active radiation (PAR^1 , MJ m⁻²d⁻¹),

¹ Abbreviations inside brackets refer to the names in the LINTUL1-program. So, e.g. I_0 is similar to PAR .

I the radiation flux that reaches the soil (PAR , $MJ\ m^{-2}d^{-1}$),

k the radiation extinction coefficient (K , $m^2\ (ground)\ m^{-2}\ (leaf)$),

L the leaf area index (LAI , $m^2\ (leaf)\ m^{-2}\ (ground)$).

The radiation extinction coefficient, k , depends on the leaf characteristics and the architecture of the crop: erect leaves intercept less radiation as compared to more horizontal leaves.

The incident radiation flux of photosynthetically active radiation, I_0 , is about 50% of the daily total radiation (DTR), so $I_0 = 0.5\ DTR$.

The amount of intercepted photosynthetically active radiation, I_{int} , is the difference between the incident radiation, I_0 , and the amount reaching the soil surface, I :

$$I_{int} = I_0 (1 - e^{-kL}) = 0.5\ DTR (1 - e^{-kL}) \quad (2)$$

Both, DTR and L in Eq. (2) vary in time and thus also the amount of intercepted radiation. Daily total radiation is a measured quantity and is contained in the weather data that are needed. For the Light INTERception and Utilization model we use daily totals that are obtained by integrating a large number of measurements taken during the day. The (green) leaf area index, L , will increase with time due to growth of new leaves, but at a certain moment, leaf senescence will be stronger and the leaf area will decrease again. Leaf area dynamics will be discussed in a later section.

Combining the actual amount of intercepted photosynthetically active radiation with the radiation use efficiency enables us to calculate the daily growth rate of a crop.

Radiation or Light use efficiency Monteith (1977) showed that a graph of crop growth rate versus the cumulative amount of absorbed photosynthetically active radiation gives a straight line, Figure 1.

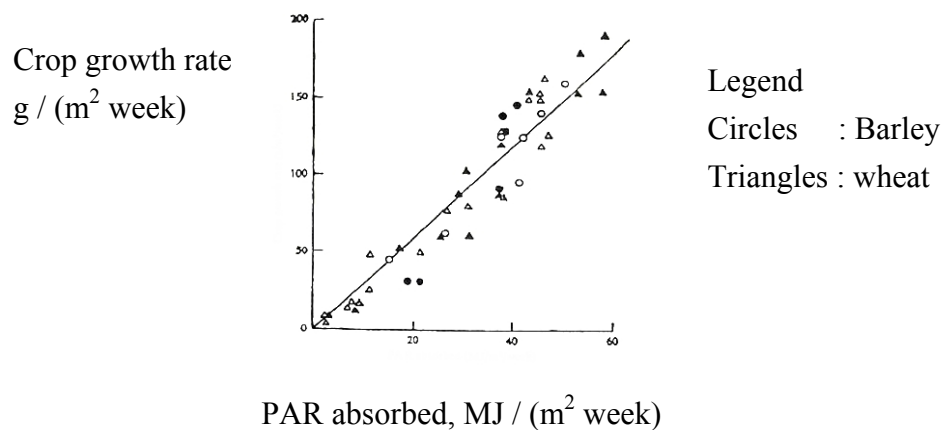


Figure 1. Crop growth rate versus the cumulative amount of absorbed photosynthetically active radiation (Gallagher and Biscoe, 1978).

The slope of this line gives an estimate of the radiation use efficiency (RUE) expressed in g dry matter per MJ of absorbed radiation. The meaning of the numerator (g dry matter) and the denominator (MJ of absorbed radiation) of this ratio is extensively discussed in the review article by Sinclair and Muchow (1999).

Crop growth rate (dW/dt ; $GTOTAL$) can now be calculated as:

$$\frac{dW}{dt} = RUE I_{int} \quad (3)$$

with the unit g (dry matter) / (m² (ground) d). The crop weight is obtained by integrating Eq. (3) over time, usually a growing season:

$$W = \int_0^t \frac{dW}{dt} dt \quad (4)$$

We need to distinguish between growth and development. Growth is determined by the product of radiation interception and the efficiency by which this radiation is used to form crop assimilates (Eq. 3). Development is determined by physiological age as discussed in the following section.

Crop development Crops pass through a number of developmental stages (*DVS*) during their live cycle. These different stages control the partitioning of assimilates or dry matter. In general, temperature has the strongest effect on development. A problem in quantifying this effect is that it is not possible to measure *development rate* directly. Instead, it must be derived from the period of time elapsed between two distinct *development stages*. The rate of development, $dDist/dt$, can then be derived using the equation:

$$\left\{ \frac{dDist}{dt} = \frac{\text{distance between 2 stages}}{\text{time period between these 2 stages}} \right\}_{T_a} \quad (5)$$

where “distance between 2 stages” may, for example, be the distance between emergence and anthesis. Such a distance could be termed 1, or 100% or so, and does not have a unit. It appears from Eq. (5) that the “time period between these 2 stages”, the denominator, in days, must be assessed for a series of constant air temperatures, T_a . Development rate is found to be roughly linearly related to temperature over a wide range, although there is usually a maximum in the response curve above which the development rate will decrease again. A graph of $dDist/dt$ versus the temperature, T , is shown in Figure 2.

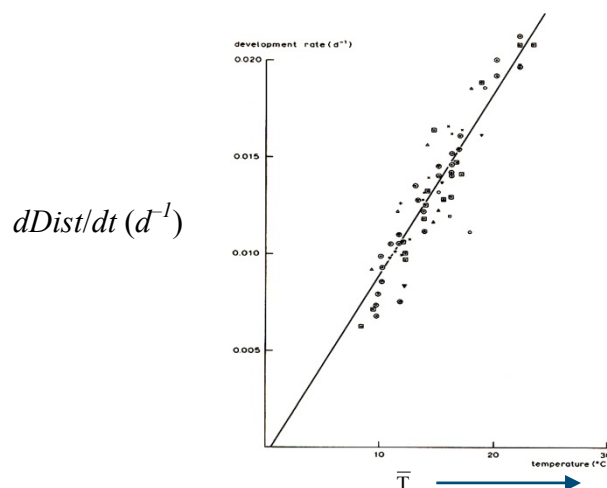


Figure 2. Development rate of spring wheat as a function of temperature from emergence till flowering (van Keulen and Seligman, 1987).

The extrapolation of the line to low temperatures yields an intercept with the temperature axis. This is often called the base temperature, T_b (*TBASE*), below which there is no or a hardly measurable development rate. Provided that air temperature, T_a (*DAVTMP*), lies between the base temperature, T_b , and the optimum temperature, beyond which the development starts to level off, the development rate is proportional to the integrated temperature

$$T_{sum} = \int (T_a - T_b) dt \quad (6)$$

From the slope of the line in Figure 2, the temperature sum, T_{sum} , to go from one development stage to the next can be calculated. The slope, α , will have the unit $1/(d \text{ } ^\circ\text{C})$, and thus $1/\alpha$ is the temperature sum.

The development rate, $dDist/dt$, is derived from experiments at constant air temperatures, T_a . In the field, however, temperatures are not constant. The temperature sum as a measure for physiological age or development stage nevertheless often works fine because of the instantaneous reaction of the crop to a change in temperature.

Allocation of assimilates In principle, photosynthesis yields assimilates that need to be partitioned over the different plant organs. In the radiation use efficiency approach, however, radiation is directly used to calculate the production of dry matter via the value of the RUE. Thus, dry matter is partitioned over the plant organs: roots, stem, leaves and storage organs. The different developmental stages control the partitioning of assimilates or dry matter. Assessing the relationships between phenological development stages and the fractions of the dry matter that are partitioned towards the different organs is still only possible by empirical work. The relationships have been assessed for a number of important crops. Figure 3 shows the data points for spring wheat.

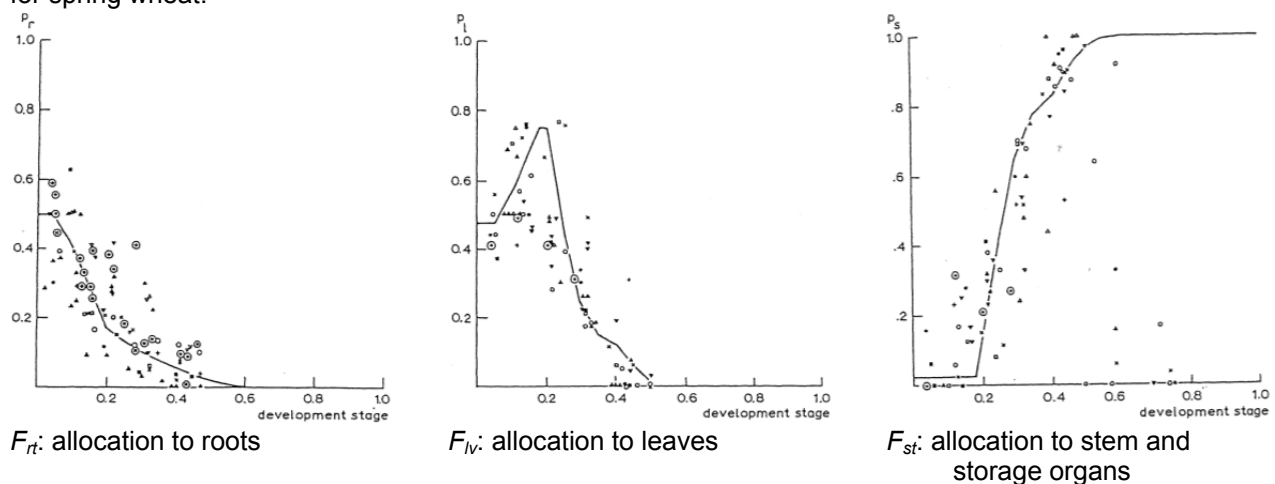


Figure 3. Allocation patterns for spring wheat (van Keulen and Seligman, 1987). Development stage one in the figures is equivalent to 2080 $^\circ\text{Cd}$. Anthesis occurs at a development stage of 0.5.

It can be seen from the figure that in the beginning much dry matter is invested in the roots and the leaves, and that after flowering, when the development stage equals 0.5, root and leaf growth stop. It appears that after $DVS=0.5$ all the dry matter is invested in the stem, but in fact the allocation to the stem is also ceasing

after $DVS=0.5$ and all the dry matter production is invested in the storage organs. Such empirical data can only be used through functions representing the partitioning fractions as a function of development stage, where the sum of the fractions should always be equal to 1.0.

The growth of each plant organ can now be calculated from Eq. (3) and the partitioning fractions depicted in Figure 3:

$$\frac{dW_i}{dt} = \frac{dW}{dt} F_i = RUE I_{int} F_i \quad (7)$$

where i represents the root (rt), stem (st), leaves (lv) or storage organ (so) of the plant, respectively. Since leaves partially senesce during the growing season, the equation for the net rate of change of green leaves, dW_{lvg}/dt ($RWLVG$), is slightly different:

$$\frac{dW_{lvg}}{dt} = \frac{dW_{lv}}{dt} - r_d W_{lv} = RUE I_{int} F_{lv} - r_d W_{lv} \quad (8)$$

where the leaf death rate is taken proportional to the green leaf weight, W_{lv} ($WLVG$, g DM m^{-2}) and a relative senescence rate, r_d (RDR , d^{-1}). The relative death rate is further detailed in the section about *Leaf area senescence*. The senescence rate $r_d W_{lv}$ in Eq. (8) is similar to DLV in the LINTUL1 program. The other organs (root, stem, storage organs) are assumed to negligibly die during the season.

Describing crop growth and dry matter allocation according to Eq. (7) implies that it is assumed that all dry matter produced in a day is directly distributed over the plant organs, and thus no reserve pool of assimilates or dry matter is considered. Redistribution of dry matter at the end of the season to the storage organs is assumed negligible.

Leaf area growth Leaves are the major photosynthesizing organs, and the calculation of intercepted photosynthetically active radiation, PAR , is largely based on their surface area. Correct simulation of the time course of the leaf area index, L (LAI), is the first requirement for correct crop growth calculations. Roughly, two different situations exist with respect to the formation of new leaf area: its growth may be *sink-limited* or it may be *source-limited*.

During juvenile growth, temperature is the overriding factor, as the rate of leaf appearance and final leaf size are constrained by temperature through its effect on cell division and extension, rather than by the supply of assimilates. In these early, sink-limited stages, leaf area increases approximately exponentially over time, Eq. (9):

$$\frac{dL}{dt} = r_g L \quad (9)$$

with r_g the relative growth rate. This relative growth rate is the product of r_l ($RGRL$), the relative growth rate of leaf area during the juvenile exponential growth phase in $(^{\circ}Cd)^{-1}$, and the effective temperature T_{eff} ($DTEFF$), which equals $(T_a - T_b)$ if $T_a \geq T_b$:

$$r_g = r_l T_{eff} \quad (10)$$

Integrating Eq. (9) and substituting Eq. (10) yields Eq. (11):

$$L = L_0 e^{(r_l T_{eff} t)} \quad (11)$$

Eq. (11) can be used to derive the exact rate of leaf area development over e.g. daily time intervals. To this purpose Eq. (11) is rewritten as

$$L_{t+\Delta t} = L_t e^{(r_l T_{eff} \Delta t)}$$

with L_t and $L_{t+\Delta t}$ the current leaf area and the leaf area at the end of a time step Δt ($DEL T$, d), respectively.

Subsequently, the slope of the exponential curve can be approximated by:

$$\frac{dL}{dt} = \frac{L_{t+\Delta t} - L_t}{\Delta t} = \frac{L_t e^{(r_l T_{eff} \Delta t)} - L_t}{\Delta t} = \frac{L_t (e^{(r_l T_{eff} \Delta t)} - 1)}{\Delta t} \quad (12)$$

Equation 12 is used in the LINTUL1 program, rather than Eq. (9), because Eq. (12) does not suffer from the accumulation of numerical integration errors.

Examination of unpublished field data suggests that a safe approximation is to restrict the exponential phase to the situation where $L < 0.75 \text{ m}^2 \text{ m}^{-2}$ and $T_{SUM} < 330 \text{ }^\circ\text{Cd}$.

In later development stages, leaf area expansion is increasingly restricted by assimilate supply (source-limited). Branching and tillering generate an increasing number of sites per plant where leaf initiation can take place and an increasing number of cells that can expand, while mutual shading of plants and leaves reduces the assimilate supply per growing point and per cell. During this stage ($L \geq 0.75 \text{ m}^2 \text{ m}^{-2}$ or $T_{SUM} \geq 330 \text{ }^\circ\text{Cd}$), the model calculates the growth of leaf area by multiplying the simulated rate of increase in leaf weight, $dW/dt F_{lv}$ (GLV, g (leaf) m^{-2} (soil) d^{-1}) by the specific leaf area of new leaves, S_{la} (SLA, m^2 (leaf) g^{-1} (leaf)), Eq. (13):

$$\frac{dL}{dt} = \frac{dW}{dt} F_{lv} S_{la} \quad (13)$$

where dW/dt is defined by Eq. (3). The dynamics of S_{la} are not well understood yet. Therefore, it is assumed constant throughout the season.

Leaf area senescence The senescence rate of L ($DLAI$, d^{-1}), is described on the basis of a relative death rate, r_d (RDR , m^2 (leaf) m^{-2} (soil) / (m^2 (leaf) m^{-2} (soil) d) or just d^{-1}), Eq. (14)

$$\frac{dL}{dt} = -r_d L \quad (14)$$

where r_d is set to the maximum of either a relative death rate due to ageing, r_{d-ag} ($RDRDV$, d^{-1}), or one due to self-shading, r_{d-sh} ($RDRSH$, d^{-1}):

$$r_d = \max(r_{d-ag}, r_{d-sh}) \quad (15)$$

with $r_{d-ag} = f(T)$ and $r_{d-sh} = f(L)$.

The $r_{d-ag} = f(T)$ is specified by the following (x,y) pairs: (-10 , 0.03) , (10 , 0.03) , (15 , 0.04) , (30 , 0.09) , (50 , 0.09), where -10, 10, 15, 30 and 50 are temperatures and the values 0.03, 0.04 and 0.09 are the corresponding relative death rates due to ageing. Death of leaves due to ageing only occurs after anthesis, as indicated by $T_{sum} \geq T_{sum-anthesis}$ ($TSUMAN$).

The $r_{d-sh} = f(L)$ equals zero for L smaller than a critical leaf area index, L_{cr} , which is set to 4 for spring wheat ($LAICR$, m^2 (leaf) m^{-2} (soil)). For leaf area index values exceeding L_{cr} , r_{d-sh} increases linearly with increasing L till a maximum value, $r_{d-shmx} = 0.03$ ($RDRSHM$, d^{-1}) at $L = 8$.

Start conditions and environmental conditions Crop growth starts with emergence, which can be triggered either by an accumulation of heat units or just by reaching a preset date. In the first case one needs to have information about the sowing date and the temperature sum from sowing to emergence. In our case we did not have such information and therefore, a preset date is taken.

For an accurate description of the establishment of small plants, much information is needed. In the case of the light interception and utilization model we have chosen to initialize leaf area as if it appears at once: leaf area at emergence amounts to $0.012 \text{ m}^2 (\text{leaf}) \text{ m}^{-2} (\text{soil})$.

Quite some other parameters are needed too; see the statements starting with the words PARAM and FUNCTION in the program listing. These values were gathered from experimental data. Furthermore, weather data are needed. The Plant Production Systems group has a large collection of daily weather data from the Netherlands, but also from a large number of other countries. For the potential growth model, we only need daily global radiation and minimum and maximum temperatures.

The model should stop making calculations at the end of the season, or when the temperature sum for maturity has been reached, ($TSUM = 2080$ for spring wheat).

Explanation of the simulation program

*Remark: The abbreviations and units used in the following explanation of LINTUL1 are given at the end of Chapter 6B. Some of the specifics of the FST language, such as INTGRL, AFGEN, INSW, etc. are also summarized there. Full details can be found in the document 'FST-Introduction' on the course-website (see link mentioned in section **Reference to literature 'Introduction to FST'** of this chapter).*

LINTUL1 is written in FST, the FORTRAN Simulation Translator (van Kraalingen *et al.*, 1994), which runs on various computer platforms, e.g. IBM-PC or compatible, and Apple-Macintosh.

The explanatory text follows as closely as possible the computer listing of the model. Each section starts with a number of lines copied from this listing. In the following text, the inevitably awkward abbreviative terminology so typical for computer modelling, is then explained. Units of all variables and data are specified.

```
DEFINE_CALL GLA (INPUT, INPUT, INPUT, INPUT, INPUT, INPUT, INPUT, INPUT, INPUT, . . .
                INPUT, INPUT,                                OUTPUT)

TITLE LINTUL1
```

FST requires, before the program starts, a definition of the calls for subroutines that are used in the program. All variables in the subroutine-call have to be defined as INPUT or OUTPUT variables. These definitions will be used by FST for sorting the program lines. After the definition of the subroutine-call the program starts with the TITLE of the program.

1. Initial conditions and run control

```
*      Initial conditions
INCON ZERO = 0.
      WLVI = LAII / SLA
```

FST requires the name of a constant as initial value for an integration. In case this integration starts at zero (e.g. summation of temperature) the initial constant `ZERO` is used.

The initial weight of leaves at emergence (`WLVI`, g (leaf) m⁻² (ground)) is calculated as the ratio of initial Leaf Area Index (`LAII`, m² (leaf) m⁻² (ground)) and the Specific Leaf Area (`SLA`, m² (leaf) g⁻¹ (leaf)).

```
*      Run control
FINISH TSUM > 2080.
```

The simulation stops if the crop is mature. This occurs if the temperature sum reaches the value 2080 °Cd.

```
TIMER STTIME = 1.; FINTIM = 200.; DELT = 1.; PRDEL = 5.
TRANSLATION_GENERAL DRIVER='EUDRIV'
PRINT LAI, WSOtha, WSO, WST, WLVI, WRT, Tsum, DAVTMP, DTR
```

Simulation may start earlier than emergence. This is specified in the `TIMER` statement (`STTIME = 1` means: 1 January). Simulation is executed with time steps of one day (`DELT=1.`), with rectilinear integration (Euler) of the rates (`DRIVER='EUDRIV'`). Output is produced every fifth day (`PRDEL=5.`). To make sure that the simulation does not continue endlessly, the finish time (`FINTIM`) is set at day 200. The simulation will thus stop either at 200 days or when the temperature sum has reached 2080 °Cd.

In the `PRINT` line any variable can be specified. Values are written to the output file at every print interval (`PRDEL`).

2. Environmental data and temperature sum

```
DYNAMIC
WEATHER WTRDIR='C:\SYS\WEATHER\'; CNTR='IT'; ISTN=3; IYEAR=1985
*      Reading weather data from weather file:
*      RDD      Daily global radiation      J/(m2*d)
*      TMMN     Daily minimum temperature   degree C
*      TMMX     Daily maximum temperature   degree C

      DTR      = RDD/1.E+6
      DAVTMP   = 0.5 * (TMMN + TMMX)
      DTEFF    = MAX ( 0., DAVTMP-TBASE )
      EMERG    = INSW(TIME-DOYEM, 0., 1.)
      Tsum     = INTGRL(ZERO, RTsum)
      RTsum    = DTEFF*EMERG
```

Actual daily total global radiation (R_{DD} , $J\ m^{-2}\ d^{-1}$) is read from the weather data file, which contains measured values for solar radiation (400 - 2000 nm) for all days of the year. R_{DD} is converted into other units by division by 10^6 , to give D_{TR} in $MJ\ m^{-2}\ d^{-1}$.

Daily maximum and minimum temperatures (T_{MMX} and T_{MMN} , respectively, $^{\circ}C$) are also read from the weather data file, containing measured values for all days of the year. $DAVTMP$ is the daily average temperature. Since many growth processes are temperature dependent above a certain threshold temperature, an effective temperature ($DTEFF$) is calculated. For spring wheat, the threshold value (T_{BASE}) is $0\ ^{\circ}C$. The variable $EMERG$ equals 0 before emergence, and 1 after emergence, when $TIME$ is equal or larger than the daynumber of emergence ($DOYEM$). For explanation of the MAX and $INSW$ functions, see the description of FST (van Kraalingen *et al.*, 1994). Note that $TIME$ may become larger than 365, in case the simulation runs from one calendar year into the next year.

Phenological development of crops is more closely related to thermal time, i.e. the accumulated number of degree-days after emergence, than to the age of the crop in days. Therefore, the model calculates the temperature sum (T_{SUM} , $^{\circ}Cd$) by accumulating the daily values of effective temperature after emergence ($R_{T_{SUM}}$, $^{\circ}C$).

3. Leaf growth and senescence

```
CALL GLA (TIME, DOYEM, DTEFF, TSUM, LAII, RGRL, DELT, SLA, LAI, GLV, ...
          GLAI)
GLV   = FLV * GTOTAL

DLAI  = LAI * RDR
RDR   = MAX(RDRDV, RDRSH)
RDRDV = INSW(TSUM-TSUMAN, 0., AFGEN(RDRT, DAVTMP))
RDRSH = LIMIT(0., RDRSHM, RDRSHM * (LAI-LAICR) / LAICR)
DLV   = WLVG * RDR

RLAI  = GLAI - DLAI
LAI   = INTGRL(ZERO, RLAI)
```

The area of green leaves is the major determinant for light interception and utilization. The leaf area index (LAI , m^2 (leaf) m^{-2} (ground)) is obtained by integrating the net result of the leaf growth rate ($GLAI$, $m^2\ m^{-2}\ d^{-1}$), and the leaf senescence rate ($DLAI$, $m^2\ m^{-2}\ d^{-1}$).

$GLAI$ is calculated, depending on the phenological development stage, in the Subroutine GLA . Before seedling emergence ($TIME < DOYEM$), $GLAI$ equals zero. At emergence, $GLAI = LAII/DELT$. After emergence, light intensity and temperature are the environmental factors influencing the rate of leaf area expansion.

During juvenile growth, temperature is the overriding factor, as the rate of leaf appearance and final leaf size are constrained by temperature through its effect on cell division and extension, rather than by the supply of assimilates. In these early stages, leaf area increases approximately exponentially over time.

Examination of unpublished field data suggests that a safe approximation is to restrict the exponential phase to the situation where $LAI < 0.75 \text{ m}^2 \text{ m}^{-2}$ and $TSUM < 330 \text{ }^\circ\text{Cd}$. This is programmed in the Subroutine GLA. Exponential leaf area development is described analytically by:

$$LAI(t+DELT) = LAI(t) * \exp(RGRL * DTEFF * DELT)$$

so that the rate of increase in leaf area during juvenile growth is:

$$\begin{aligned} GLAI &= (LAI(t + DELT) - LAI(t)) / DELT \\ &= LAI(t) * (\exp(RGRL * DTEFF * DELT) - 1.) / DELT \end{aligned}$$

in which $LAI(t)$ is the current leaf area ($\text{m}^2 \text{ m}^{-2}$), $RGRL$ is the relative growth rate of leaf area, expressed per degree-day ($^\circ\text{Cd}^{-1}$), $DELT$ is the time step of integration (d) and $DTEFF$ is the daily effective temperature ($^\circ\text{C}$).

In later development stages, leaf area expansion is increasingly restricted by assimilate supply. Branching and tillering generate an increasing number of sites per plant where leaf initiation can take place and mutual shading of plants further reduces the assimilate supply per growing point. During this stage ($LAI > 0.75 \text{ m}^2 \text{ m}^{-2}$ or $TSUM > 330 \text{ }^\circ\text{Cd}$), the model calculates the growth of leaf area by multiplying the simulated increase in leaf weight ($GLV, \text{g m}^{-2} \text{ d}^{-1}$) by the specific leaf area of new leaves ($SLA, \text{m}^2 \text{ g}^{-1}$).

The senescence rate of LAI ($DLAI, \text{d}^{-1}$) is described on the basis of a relative death rate ($RDR, (\text{m}^2 \text{ m}^{-2})/(\text{m}^2 \text{ m}^{-2}) \text{ d}^{-1}$), set at the maximum of either a relative death rate due to ageing ($RDRDV$) or one due to self-shading, $RDRSH$. The latter equals zero for LAI smaller than 4 ($= LAICR$), and above that value increases linearly with increasing LAI till a maximum value of 0.03 ($= RDRSHM$) at $LAI = 8$. For the meaning of the $LIMIT$ function, see the description of FST, van Kraalingen *et al.*, 1994.

$RDRDV$ equals zero as long as $TSUM < TSUMAN$ (pre-anthesis stage) and is a function of the average daily temperature ($DAVTMP, ^\circ\text{C}$) for $TSUM > TSUMAN$.

The death rate of leaves in terms of weight ($DLV, \text{g m}^{-2} \text{ d}^{-1}$) is defined using the same relative senescence rate (RDR) that also applies to LAI , but now multiplied with the weight of the green leaves ($WLVG, \text{g m}^{-2}$).

4. Light interception and total crop growth rate

$$\begin{aligned} PARINT &= 0.5 * DTR * (1. - \exp(-K*LAI)) \\ GTOTAL &= LUE * PARINT \end{aligned}$$

Photosynthetically active radiation (PAR), wavelengths between 400 and 700 nm, is about 50% of the incoming global radiation ($0.5*DTR$). The daily values of intercepted photosynthetically active radiation ($PARINT, \text{MJ m}^{-2} \text{ d}^{-1}$) are derived by assuming that light interception increases with LAI according to a negative exponential function of leaf area index, characterized by a crop-specific light extinction coefficient ($K, \text{m}^2 (\text{ground}) \text{ m}^{-2} (\text{leaf})$).

The overall daily growth rate of the crop ($GTOTAL, \text{g m}^{-2} \text{ d}^{-1}$) is then calculated by multiplying the amount of light intercepted by a constant light use efficiency ($LUE, \text{g MJ}^{-1}$).

5. Growth rates and dry matter production of plant organs

```

FRT    = AFGEN( FRTTB, TSUM )
FLV    = AFGEN( FLVTB, TSUM )
FST    = AFGEN( FSTTB, TSUM )
FSO    = AFGEN( FSOTB, TSUM )

```

```

WLVG   = INTGRL( WLVI, RWLVG)
WLVD   = INTGRL( ZERO, DLV  )
WST    = INTGRL( ZERO, RWST )
WSO    = INTGRL( ZERO, RWSO )
WSOtha = WSO / 100.
WRT    = INTGRL( ZERO, RWRT )
WLV    = WLVG + WLVD
RWLVG  = GTOTAL * FLV - DLV
RWST   = GTOTAL * FST
RWSO   = GTOTAL * FSO
RWRT   = GTOTAL * FRT

```

Partitioning of biomass over the various plant organs is described by fixed distribution factors, defined as functions of the temperature sum. Before anthesis ($TSUM < TSUMAN$) the highest distribution factors are those for roots (FRT), leaves (FLV) and stems (FST), thereafter most of the biomass is allocated to the storage organs, i.e. grains (FSO). This allocation-pattern is embodied in the FUNCTION-statements given in the next section.

Dry weights of the various plant organs (roots (WRT , $g\ m^{-2}$), green leaves ($WLVG$, $g\ m^{-2}$), dead leaves ($WLVD$, $g\ m^{-2}$), stems (WST , $g\ m^{-2}$), storage organs (WSO , $g\ m^{-2}$)) are obtained through integration of the respective growth rates. For convenience, the program calculates yield in tons per ha as well ($WSOtha$, $t\ ha^{-1}$).

6. Functions and parameters for spring wheat

```

*      Section 1
PARAM LAII = 0.012; SLA = 0.022
*      Section 2
PARAM TBASE = 0.
*      Section 3
PARAM DOYEM = 32.
PARAM RGRL = 0.009; LAICR = 4.; TSUMAN = 1110. ; RDRSHM = 0.03
FUNCTION RDRT = -10.,0.03, 10.,0.03, 15.,0.04, 30.,0.09, 50.,0.09
*      Section 4
PARAM LUE = 3.0; K = 0.6
*      Section 5
*      Partitioning tables for leaves (LV), stems (ST),
*      storage organs (SO) and roots (RT):
FUNCTION FRTTB = 0.,0.50, 110.,0.50, 275.,0.34, 555.,0.12, ...

```

```

      780.,0.07, 1055.,0.03, 1160.,0.02, 1305.,0. , 2500.,0.
FUNCTION FLVTB = 0.,0.33, 110.,0.33, 275.,0.46, 555.,0.44, ...
      780.,0.14, 1055.,0. , 2500.,0.
FUNCTION FSTTB = 0.,0.17, 110.,0.17, 275.,0.20, 555.,0.44, ...
      780.,0.79, 1055.,0.97, 1160.,0. , 2500.,0.
FUNCTION FSOTB = 0.,0. , ...
                  1055.,0. , 1160.,0.98, 1305.,1. , 2500.,1.
END
* Start of rerun section

PARAM DOYEM = 60.
END

STOP

* Start of Subroutine section (see Section 3)

* -----*
*SUBROUTINE GLA*
*Purpose:This subroutine computes daily increase of leaf area index*
*      (m2 leaf/ m2 ground/ d)*
* -----*
      SUBROUTINE GLA(TIME,DOYEM,DTEFF,TSUM,LAI,LAII,RGRL,DELT,SLA,LAI,GLV,
$      GLAI)
      IMPLICIT REAL (A-Z)

*---- Growth during maturation stage:
      GLAI = SLA * GLV

*---- Growth during juvenile stage:
      IF ((TSUM.LT.330.).AND.(LAI.LT.0.75))
$      GLAI = LAI * (EXP(RGRL * DTEFF * DELT) - 1.) / DELT

*---- Growth at day of seedling emergence:
      IF ((TIME.GE.DOYEM).AND.(LAI.EQ.0.))
$      GLAI = LAII / DELT

*---- Growth before seedling emergence:
      IF (TIME.LT.DOYEM) GLAI = 0.

      RETURN
      END

```

REFERENCE TO LITERATURE 'INTRODUCTION TO FST'

For those who want to know more about FST-program an introductory reader will be published on the BlackBoard course-site ([see MyPortal >>> link to HPC23303 BlackBoard-site](#))

FST, the Fortran Simulation Translator

Authors: D.W.G. van Kraalingen, C. Rappoldt and H.H. van Laar

Teacher: A.G.T. Schut

Text from the book:

On systems analysis and simulation of ecological processes; with examples in CSMP, FST and FORTRAN. Second Edition. P.A. Leffelaar, 1999 (Ed.).

Series title: Current Issues in Production Ecology, Vol 4. Kluwer academic publishers, 318 pp.

EXERCISES WITH RESULTS FROM SIMULATIONS WITH LINTUL-1 (BRAZIL)

Purpose: to get a feeling about the order of magnitude of potential crop production under different climatic conditions, and crop-model sensitivity for emergence date and some parameters.

*Note: choose either wheat growth in **Brazil** or in the **Netherlands**. If you are fast enough you can also compare the results of the two locations.*

The first part is prepared for those who wish to study potential production for **Brazil**: (p. 14-16)

The effect of weather and different emerging dates on crop growth

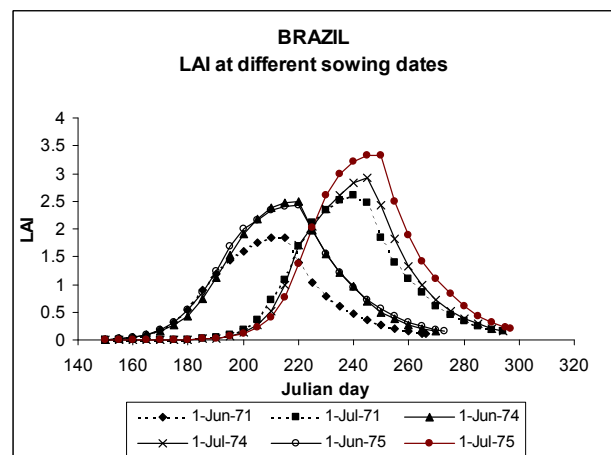
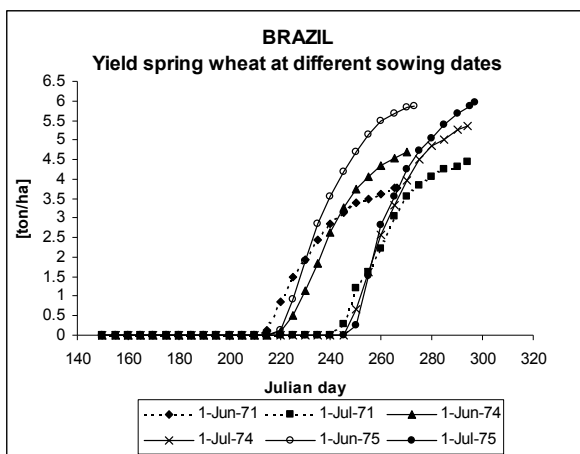
Crop growth is strongly dependent on the weather. Variables that occur in the weather files are: day of the year; the irradiation ($\text{kJ m}^{-2} \text{d}^{-1}$); minimum temperature ($^{\circ}\text{C}$); maximum temperature ($^{\circ}\text{C}$); vapour pressure (kPa); mean wind speed (m s^{-1}); precipitation (mm d^{-1}).

We give results for the years 1971 (dry), 1974 (average year), or 1975 (wet). Furthermore, we have generated two data sets with LINTUL1 for each of these years:

Run 1: Emergence at 1 June (that is the default value: DOYEM=152.);

Run 2: Emergence at 1 July (DOYEM=182.);

Results are given for the yields, left figure, and for the LAI, right figure:



The numerical values of yield results and the difference between the yields for the different sowing dates are also given:

- 1971 (dry)

difference emerg. June 1 (152) -	emerg. July 1 (182)	
3.78 -	4.44	= 0.66 t/ha (= 66 g.m^{-2})

big difference in LAI development, see above figures of LAI versus time.
- 1974 (average)

difference	4.70 -	5.35	
			= 0.65 t/ha (= 65 g.m^{-2})

big difference in LAI, and also larger than dry year 1971
- 1975 (wet)

difference	5.88 -	5.95	
			= 0.07 t/ha (= 7 g.m^{-2})

big difference in LAI, and also little larger than 1974 (aver.)

Questions:

- 1) Which weather variables are important for calculating potential crop growth?
- 2) In which type of year is the yield highest.
- 3) Explain the possible causes why in that type of year the yield is highest, using the information of both figures.
- 4) Which sowing date would you recommend to a farmer if these figures were confirmed by experimental data?

The sensitivity of yield for some important crop parameters

The light extinction coefficient (K), the specific leaf area (SLA), the initial relative growth rate of the leaves (RGRL), the day of emergence (DOYEM), and the Light Use Efficiency (LUE) all affect crop yield. The Lintul-1 crop growth model was used to see which of these variables appear most important in determining yield. A model is said to be sensitive for a parameter if a slight change in that parameter causes an appreciable change in the model result. The results were all compared to the standard results, the so called default parameters.

We give the results for six model runs for each year (dry, average, wet), so in total 18 runs:

Run 0: Standard run with all parameters at the default value as indicated in the listing

Run 1: Increase K by 10%

Run 2: Increase SLA by 10%

Run 3: Increase RGRL by 10%

Run 4: Delay sowing, i.e. emergence day DOYEM, by 10 d.

Run 5: Increase LUE by 10%

The changes made in run 1-3 all increase cumulative light interception by the crop. The change in LUE, run 5, increases the efficiency by which the light is utilized to form dry matter in the crop.

In summary, the parameters are adapted as follows:

Default values

K = 0.6

SLA = 0.022

RGRL = 0.009

DOYEM 10 days later, so from 152 to →

LUE = 3.0

Adapted values:

Run	1	+10% = 0.66
	2	+10% = 0.0242
	3	+10% = 0.0099
	4	+10 d = 162
	5	+10% = 3.3

Our results are given below:

- 1971 dry

default yield (=Doyem 152) = 3.78

	ΔK	ΔSLA	$\Delta RGRL$	$\Delta DOYEM$	ΔLUE
	4.42	4.23	4.34	4.32	4.66
diff.	0.64	0.45	0.56	0.54	0.88

- 1974 average

default yield = 4.70

	ΔK	ΔSLA	$\Delta RGRL$	$\Delta DOYEM$	ΔLUE
	5.42	5.25	5.25	5.98	5.77
diff.	0.72	0.55	0.55	1.28	1.07

- 1975 wet

default yield = 5.88

	ΔK	ΔSLA	$\Delta RGRL$	$\Delta DOYEM$	ΔLUE
	6.75	6.53	6.53	6.31	7.19
diff.	0.87	0.65	0.65	0.43	1.31

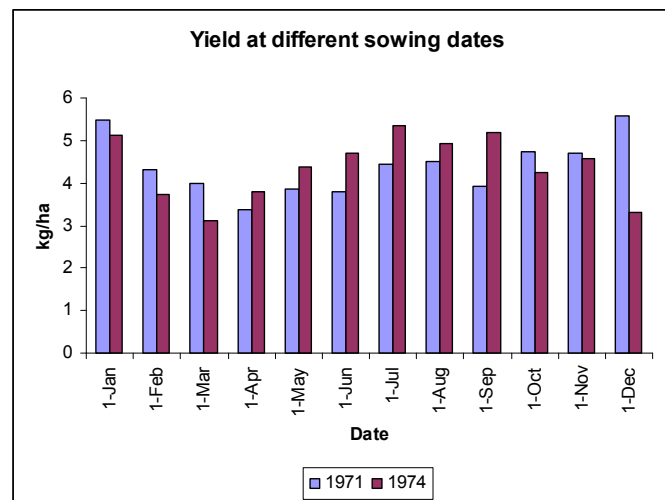
Questions:

- 5) What are the units of K, SLA, RGRL, DOYEM and LUE. Include in these units also the biological or physical entity. For example not only g, but g biomass, not g/g but, for instance, g water / g soil (soil moisture content on a weight basis).
- 6) Which parameter change increases final yield most? Distinguish between the parameters affecting the cumulative light interception of the crop (light capture) and the light utilization. Explain the reasons for this.

The effect of sowing date on crop growth

The effect of sowing date on crop performance can easily be investigated by a model, rather than in experiments, although that might be a very interesting exercise, of course. To this purpose we made twelve runs and varied the day of emergence by setting DOYEM at 1 January, 1 February, 1 March ... up to 1 December. Because no weather for 1976 is available, not all runs can be made for the wet year. They are therefore omitted here.

Our results are as follows:



Questions:

- 7) Which month gives the highest potential yield?
- 8) Comment on the assumption that emergence and growth takes place throughout the whole of the year.

EXERCISES WITH RESULTS FROM SIMULATIONS WITH LINTUL-1 (NL)

Purpose: to get a feeling about the order of magnitude of potential crop production under different climatic conditions, and crop-model sensitivity for emergence date and some parameters.

*Note: choose either wheat growth in **Brazil** or in the **Netherlands**. If you are fast enough you can also compare the results of the two locations.*

This part is for those who wish to study potential production for the **Netherlands**: (p. 17-19)

The effect of weather and different emerging dates on crop growth

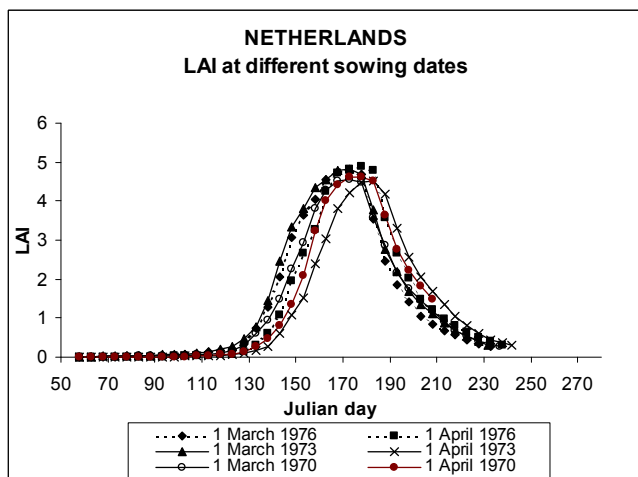
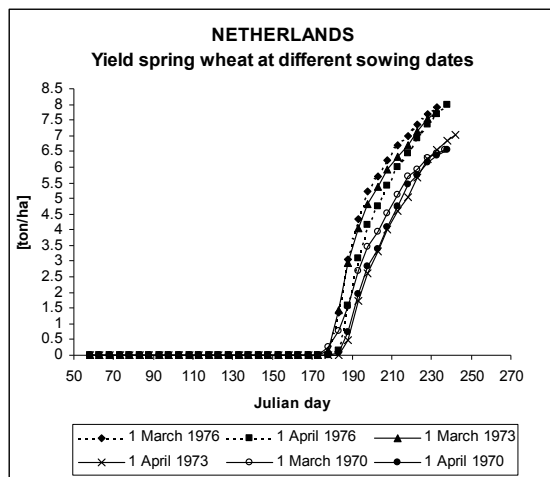
Crop growth is strongly dependent on the weather. Variables that occur in the weather files are: day of the year; the irradiation ($\text{kJ m}^{-2} \text{d}^{-1}$); minimum temperature ($^{\circ}\text{C}$); maximum temperature ($^{\circ}\text{C}$); vapour pressure (kPa); mean wind speed (m s^{-1}); precipitation (mm d^{-1}).

We give results for the years 1976 (dry), 1973 (average year), or 1970 (wet). Furthermore, we have generated two data sets with LINTUL1 for each of these years:

Run 1: Emergence at 1 March (that is the default value: DOYEM=60.);

Run 2: Emergence at 1 April (DOYEM=91.);

Results are given for the yields, left figure, and for the LAI, right figure:



The numerical values of yield results and the difference between the yields for the different sowing dates are also given:

- 1970 (wet)

difference emerg. March 1 (60) -	emerg. April 1 (91)	
6.56	-	6.76 = 0.20 t/ha

small difference in LAI's, see the above figures of LAI versus time.
- 1973 (average)

difference	7.72	-	7.01	= 0.71 t/ha
------------	------	---	------	-------------

also little difference in LAI
- 1976 (dry)

difference	7.92	-	7.99	= 0.07 t/ha
------------	------	---	------	-------------

figure LAI about similar to 1970

Questions:

- 1) Which weather variables are important for calculating potential crop growth?
- 2) In which type of year is the yield highest.
- 3) Explain the possible causes why in that type of year the yield is highest, using the information of both figures.
- 4) Which sowing date would you recommend to a farmer if these figures were confirmed by experimental data?

The sensitivity of yield for some important crop parameters

The light extinction coefficient (K), the specific leaf area (SLA), the initial relative growth rate of the leaves (RGRL), the day of emergence (DOYEM), and the Light Use Efficiency (LUE) all affect crop yield. The Lintul-1 crop growth model was used to see which of these variables appear most important in determining yield. A model is said to be sensitive for a parameter if a slight change in that parameter causes an appreciable change in the model result. The results were all compared to the standard results, the so called default parameters.

We give the results for six model runs for each year (dry, average, wet), so in total 18 runs:

Run 0: Standard run with all parameters at the default value as indicated in the listing

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The changes made in run 1-3 all increase cumulative light interception by the crop. The change in LUE, run 5, increases the efficiency by which the light is utilized to form dry matter in the crop.

In summary, the parameters are adapted as follows:

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LUE = 3.0

Adapted values:

Run	1	+10%	= 0.66
	2	+10%	= 0.0242
	3	+10%	= 0.0099
	4	+10 d	= 162
	5	+10%	= 3.3

Our results are given below:

- 1970 wet

default yield (=Doyem 60)=6.56

	ΔK	ΔSLA	$\Delta RGRL$	$\Delta DOYEM$	ΔLUE
	7.05	6.82	6.76	6.50	7.50
diff.	0.49	0.26	0.20	-0.06	0.94

- 1973 average

default yield =7.72

	ΔK	ΔSLA	$\Delta RGRL$	$\Delta DOYEM$	ΔLUE
	8.19	7.94	7.86	7.53	8.73
diff.	0.47	0.22	0.14	-0.19	1.01

- 1976 dry

default yield =7.92

	ΔK	ΔSLA	$\Delta RGRL$	$\Delta DOYEM$	ΔLUE
	8.47	8.19	8.10	7.95	9.01
diff.	0.55	0.27	0.18	0.03	1.09

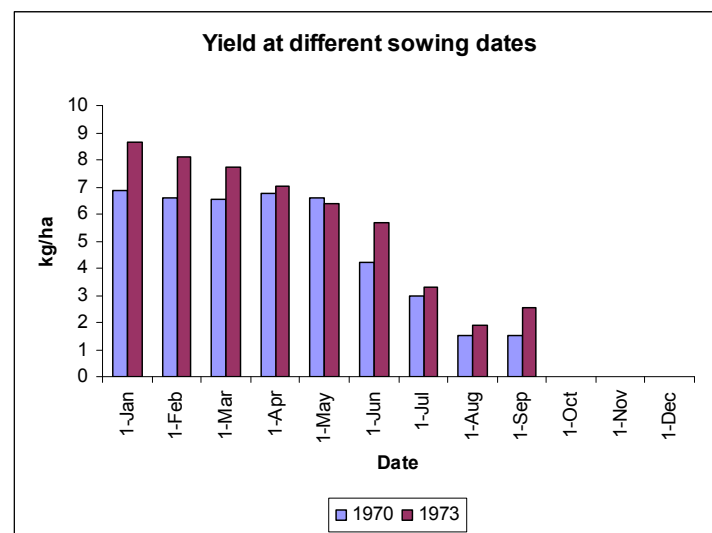
Questions:

- 5) What are the units of K, SLA, RGRL, DOYEM and LUE. Include in these units also the biological or physical entity. For example not only g, but g biomass, not g/g but, for instance, g water / g soil (soil moisture content on a weight basis).
- 6) Which parameter change increases final yield most? Distinguish between the parameters affecting the cumulative light interception of the crop (light capture) and the light utilization. Explain the reasons for this.

The effect of sowing date on crop growth

The effect of sowing date on crop performance can easily be investigated by a model, rather than in experiments, although that might be a very interesting exercise, of course. To this purpose we made twelve runs and varied the day of emergence by setting DOYEM at 1 January, 1 February, 1 March ... up to 1 December. Because no weather for 1977 is available, not all runs can be made for the dry year. They are therefore omitted here.

Our results are as follows:



Questions:

- 7) Which month gives the highest potential yield?
- 8) Comment on the assumption that emergence and growth takes place throughout the whole of the year.

TABLE WITH DAYNUMBER OF YEAR (DOY):

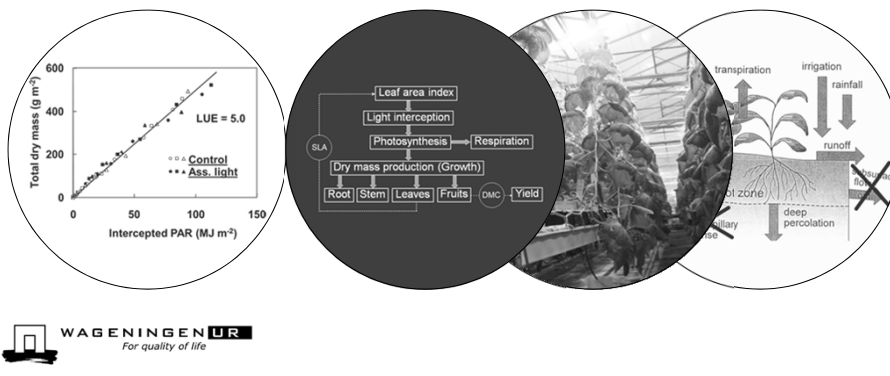
DAYNUMBER OF YEAR (DOY)

JAN	FEB	MAR	APRIL	MAY	JUNE	JULY	AUG	SEP	OCT	NOV	DEC	DAY
1	32	60	91	121	152	182	213	244	274	305	335	1
2	33	61	92	122	153	183	214	245	275	306	336	2
3	34	62	93	123	154	184	215	246	276	307	337	3
4	35	63	94	124	155	185	216	247	277	308	338	4
5	36	64	95	125	156	186	217	248	278	309	339	5
6	37	65	96	126	157	187	218	249	279	310	340	6
7	38	66	97	127	158	188	219	250	280	311	341	7
8	39	67	98	128	159	189	220	251	281	312	342	8
9	40	68	99	129	160	190	221	252	282	313	343	9
10	41	69	100	130	161	191	222	253	283	314	344	10
11	42	70	101	131	162	192	223	254	284	315	345	11
12	43	71	102	132	163	193	224	255	285	316	346	12
13	44	72	103	133	164	194	225	256	286	317	347	13
14	45	73	104	133	165	195	226	257	287	318	348	14
15	46	74	105	135	166	196	227	258	288	319	349	15
16	47	75	106	136	167	197	228	259	289	320	350	16
17	48	76	107	137	168	198	229	260	290	321	351	17
18	49	77	108	138	169	199	230	261	291	322	352	18
19	50	78	109	139	170	200	231	262	292	323	353	19
20	51	79	110	140	171	201	232	263	293	324	354	20
21	52	80	111	141	172	202	233	264	294	325	355	21
22	53	81	112	142	173	203	234	265	295	326	356	22
23	54	82	113	143	174	204	235	266	296	327	357	23
24	55	83	114	144	175	205	236	267	297	328	358	24
25	56	84	115	145	176	206	237	268	298	329	359	25
26	57	85	116	146	177	207	238	269	299	330	360	26
27	58	86	117	147	178	208	239	270	300	331	361	27
28	59	87	118	148	179	209	240	271	301	332	362	28
29		88	119	149	180	210	241	272	302	333	363	29
30		89	120	150	181	211	242	273	303	334	364	30
31		90		151		212	243		304		365	31

HANDOUTS: LINTUL1 - POTENTIAL CROP GROWTH

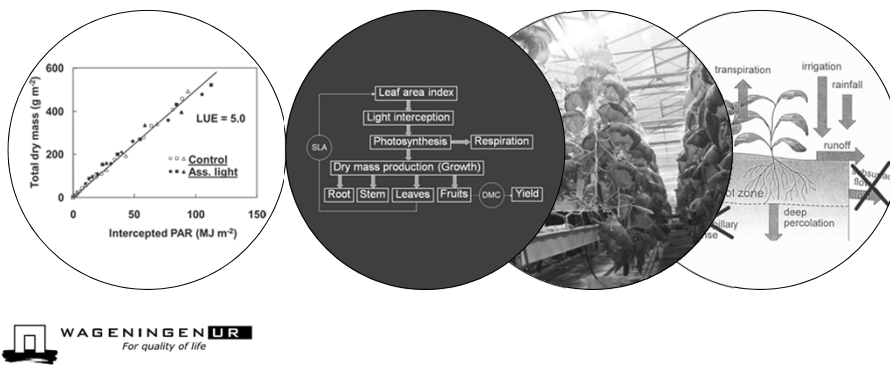
Lintul1: Potential crop growth

Dr. ir. Peter Leffelaar - Wageningen University

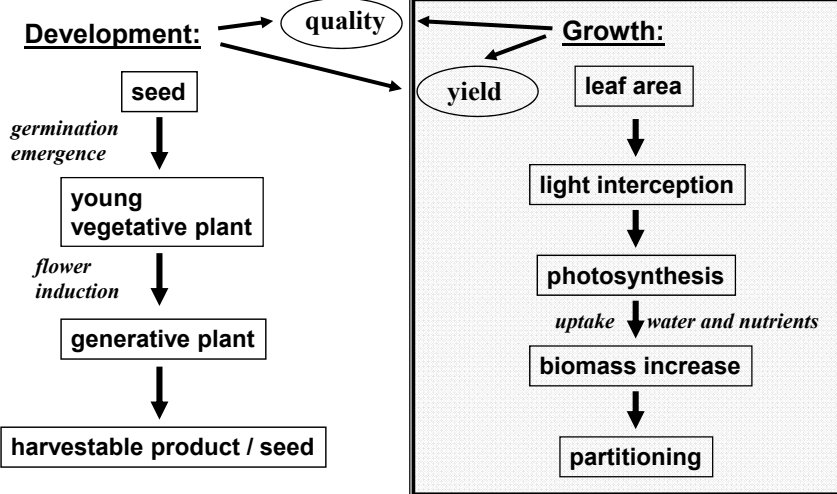


Lintul1: Potential crop growth

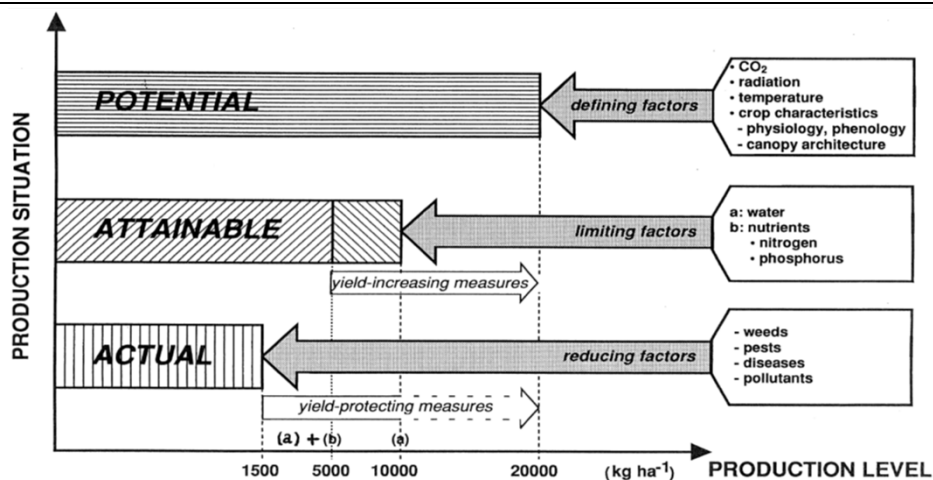
Introduction



Lintul1: Potential crop growth - scope of the course -

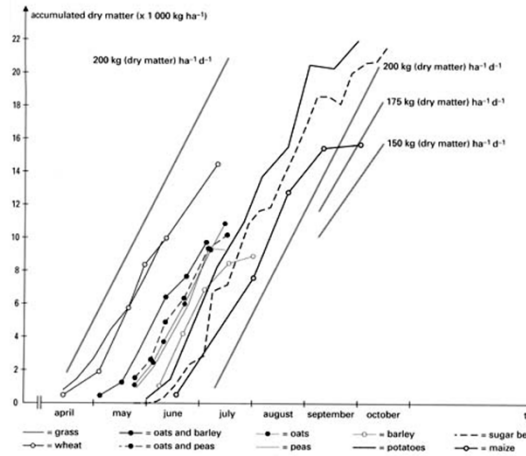


Potential, attainable and actual production levels



Schematic overview of production situations with corresponding dry matter production levels

Optimal growth rates - rule of thumb



Growth rates of the main agricultural crops in the Netherlands under (near)-optimal growth conditions, compared to growth rates of 200, 175 and 150 kg (dry matter) ha⁻¹ d⁻¹
 (Source: Sibma, 1968)

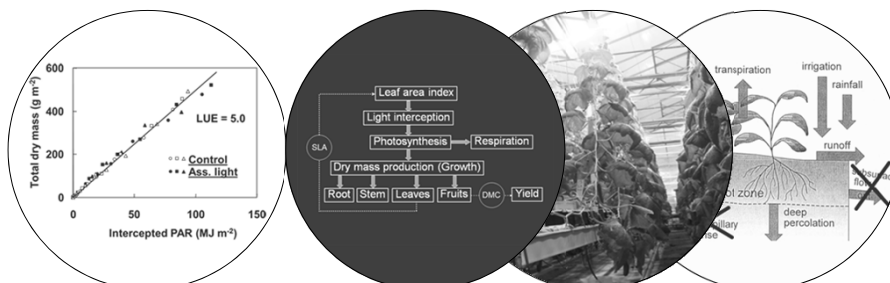
From: Sibma, 1968



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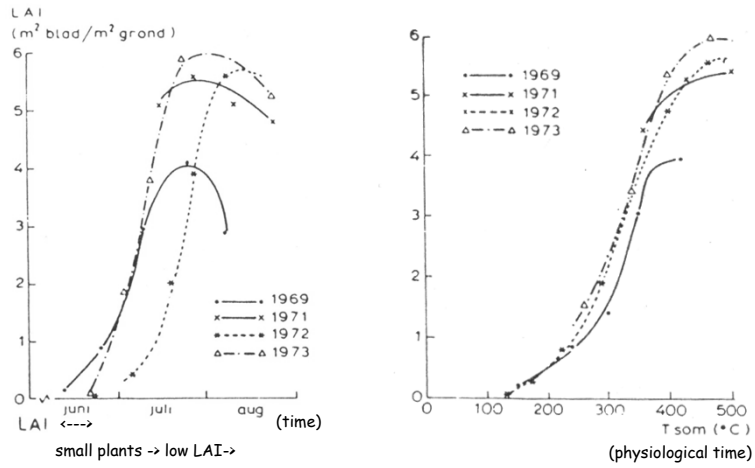
Lintul1: Potential crop growth

Light and light interception



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 For quality of life

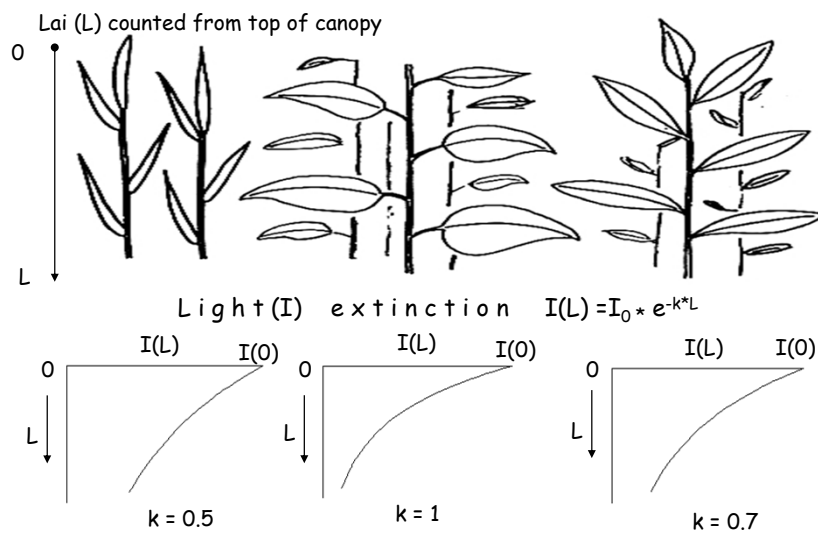
Leaf area of maize as a function of (physiological) time



small plants → low LAI →
not all sunlight used →
growth rate << 200 kg/(ha*d)

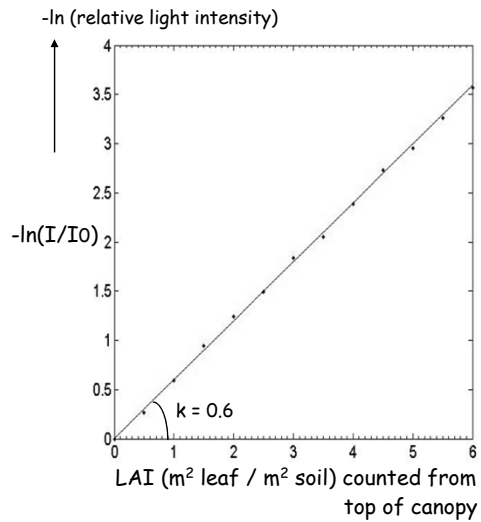
From: Sibma, 1987

Light extinction is a function of LAI



From: Coombs et al., 1985

Light extinction coefficient, k , follows from slope



$$I(L) = I_0 \cdot e^{-k \cdot L}$$

$$\Rightarrow -\ln\left(\frac{I}{I_0}\right) = k \cdot L$$

maize

From: Sibma, 1987



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Spectral composition of global radiation

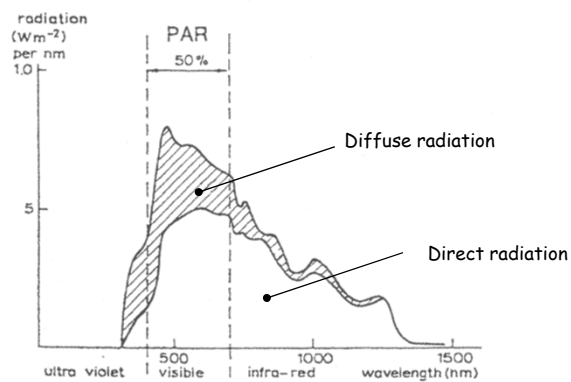


figure 2.2 Spectral composition of global radiation, discriminating between direct radiation (clear area) and diffuse (i.e. indirect) radiation scattered by clouds (hatched area). PAR = photosynthetically active radiation.
(Source: Monteith and Unsworth, 1990, Principles of environmental physics, Edward Arnold, London, 291 p.)

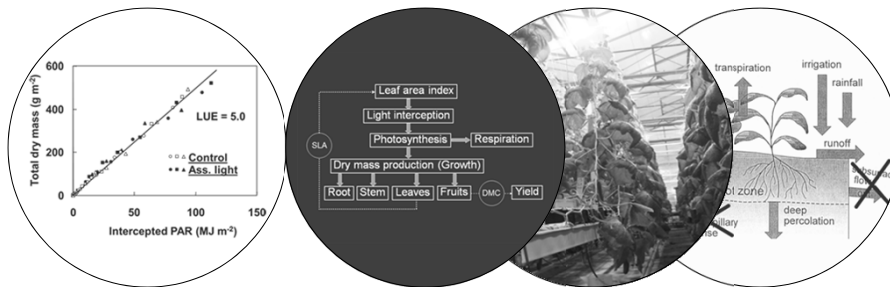
From: Monteith & Unsworth, 1990



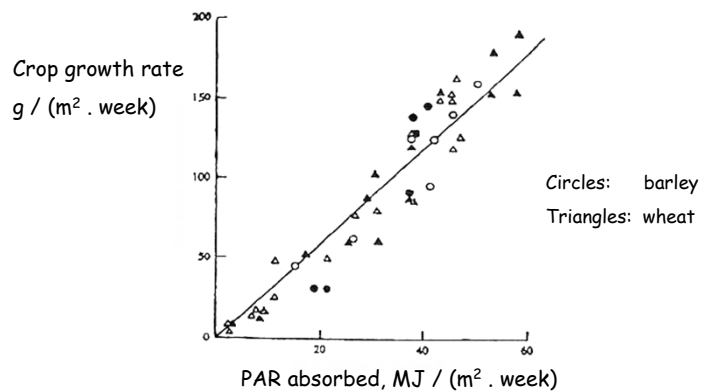
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Lintul1: Potential crop growth

Light (or Radiation) use efficiency



Light Use Efficiency, LUE, follows from slope



PARAM LUE = 3.0 (g Dry Matter per MJ PAR)

LUE Pinus = 1.9

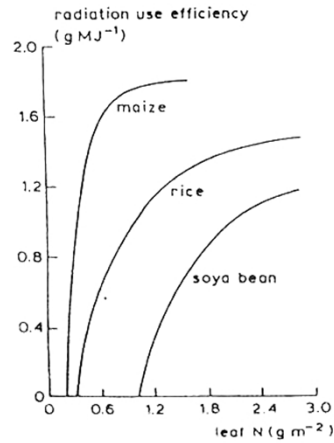
LUE Quercus = 2.1

From: Gallagher & Biscoe, 1978



See for an excellent discussion about LUE:
Sinclair and Muchow, 1999.

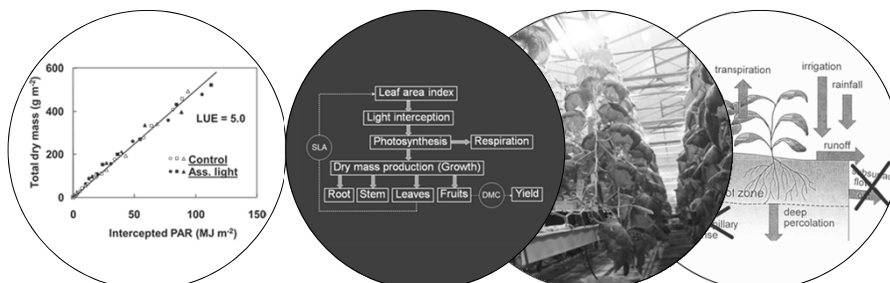
Radiation use efficiency as a function of leaf nitrogen content per unit area



From: Sinclair & Horie, 1989

Lintul1: Potential crop growth

Light interception, utilization and growth



Light interception, utilization and growth in LINTUL

TITLE LINTUL

INITIAL

INCON WTOTI = 5.

TIMER STTIME=1. ; FINTIM=100. ; DELT=1. ; PRDEL = 5.

TRANSLATION_GENERAL DRIVER = 'EUDRIV'

PRINT LAI, WTHA, PARINT

DYNAMIC

LAI = WTOTAL * LWR * SLA

PARINT = (0.5 * DTR) * (1. - EXP(-K * LAI))

$L = W F_{lv} S_{la}$

$I_{int} = 0.5 DTR (1 - e^{-kL})$

compare Eq.(13)

Eq.(2)

GTOTAL = LUE * PARINT

WTOTAL = INTGRL(WTOTI, GTOTAL)

WTHA = WTOTAL / 100.

$dW/dt = RUE I_{int}$

$W = \int dW/dt dt$

Eq.(3)

Eq.(4)

PARAM DTR = 20. ; LWR = 0.5 ; SLA = 0.022 ; LUE = 3.0 ; K = 0.6

END

STOP

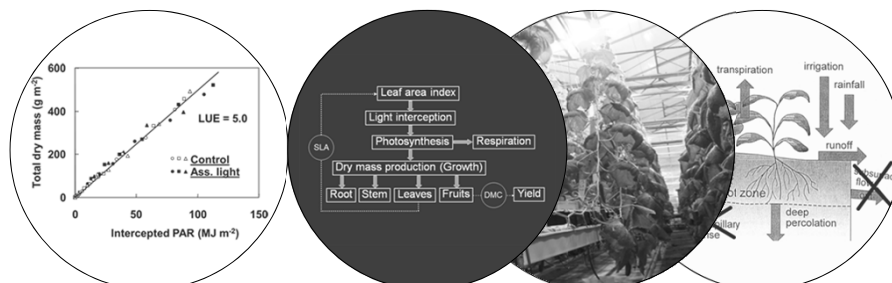
ENDJOB



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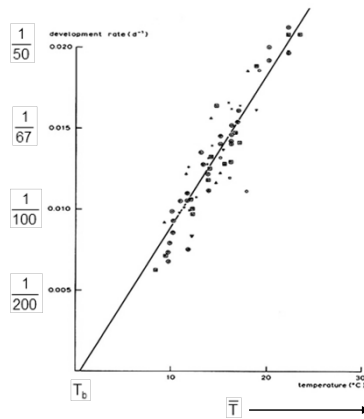
Lintul1: Potential crop growth

Crop development



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For quality of life

Development rate of spring wheat as a function of temperature from emergence till flowering . . .



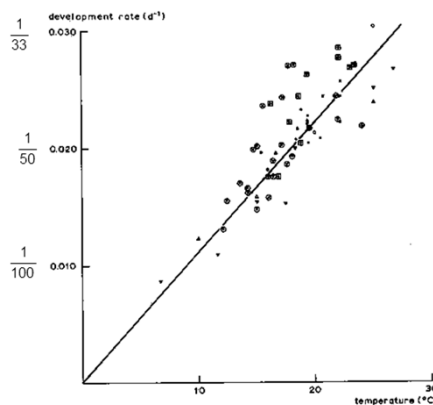
$$\frac{1}{\# \text{ days}} = \bar{T} \cdot 0.0009 \Rightarrow$$

$$\text{"temperature sum"} = \bar{T} \cdot \# \text{ days} \Rightarrow$$

$$\frac{1}{0.0009} = 1110 \text{ }^{\circ}\text{Cd}$$

Data from over 20 papers assembled
by: Van Keulen & Seligman, 1987

. . . and from flowering to ripeness



$$\frac{1}{\# \text{ days}} = \bar{T} \cdot 0.00103 \Rightarrow$$

$$\bar{T} \cdot \# \text{ days} = \frac{1}{0.00103} = 970 \text{ }^{\circ}\text{Cd}$$

Vegetative Phase: 1110 °Cd
Generative Phase: 970 °Cd
Whole growing season: 2080 °Cd

finish condition Lintul

Data from over 20 papers assembled
by: Van Keulen & Seligman, 1987

Use of weather data in LINTUL

TITLE LINTUL

* 1. Initial conditions and run control

FINISH TSUM > 2080.

* 2. Environmental data and temperature sum

WEATHER WTRDIR='C:\SYS\WEATHER'; CNTR='IT'; ISTN=3; ...
IYEAR=1985

DTR = RDD / 1.E+6

DAVTMP = 0.5 * (TMMN + TMMX)

DTEFF = MAX(0. , DAVTMP - TBASE)

EMERG = INSW(TIME-DOYEM , 0. , 1.)

TSUM = INTGRL (ZERO, RTSUM)

RTSUM = DTEFF * EMERG

$$T_{sum} = \int (T_a - T_b) dt$$

* 6. Functions and parameters for spring wheat

PARAM TBASE = 0.

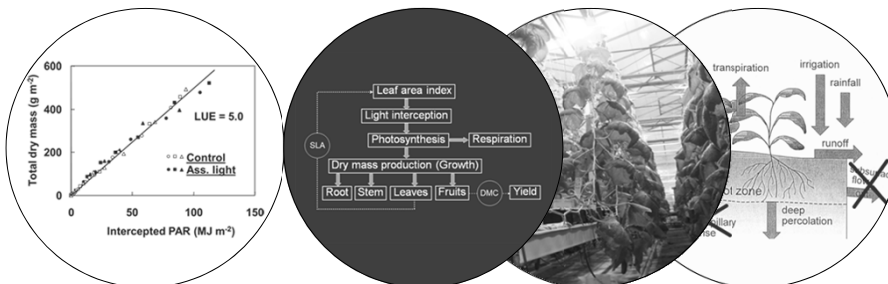
PARAM DOYEM = 32.



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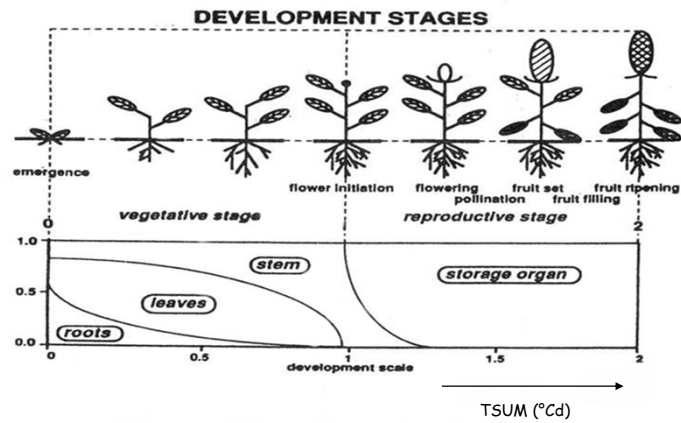
Lintul1: Potential crop growth

Development stages and allocation of dry matter



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Crop development stages

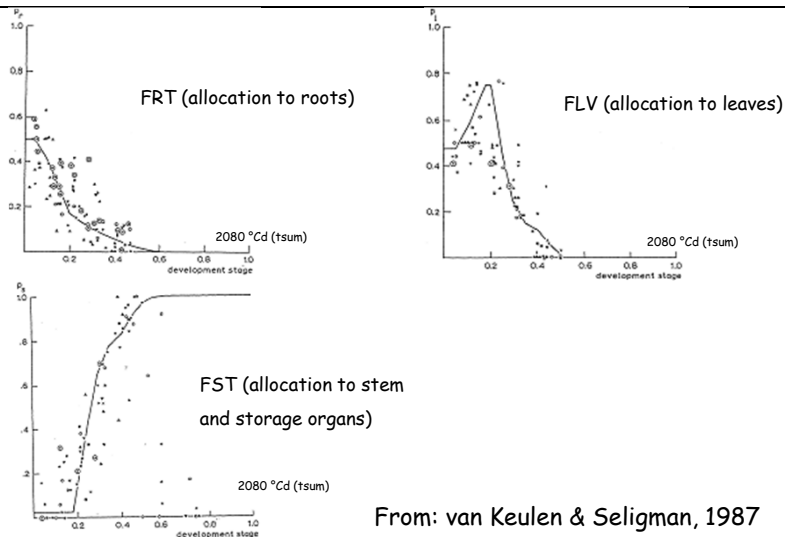


From: L6venstein, Lantinga, Rabbinge and van Keulen, 1995



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Substrate allocation patterns in Spring Wheat: Measured

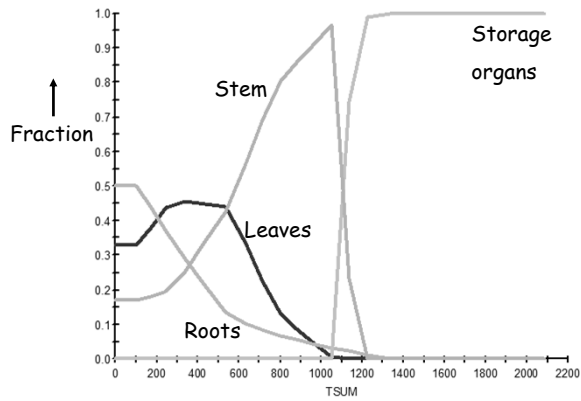


From: van Keulen & Seligman, 1987



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Substrate allocation patterns in Spring Wheat: In model



Allocation pattern of assimilates for spring wheat
as used in Lintul (Light interception and utilization)



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Growth rates and dry matter production of plant organs

TITLE LINTUL

FRT = AFGEN(FRTTB , Tsum)
FLV = AFGEN(FLVTB , Tsum)
FST = AFGEN(FSTTB , Tsum)
FSO = AFGEN(FSOTB , Tsum)

WLVG = INTGRL(WLVI , RWLVG)
WLVD = INTGRL(ZERO , DLV)
WST = INTGRL(ZERO , RWST)
WSO = INTGRL(ZERO , RWSO)
WSO_{THA} = WSO / 100.
WRT = INTGRL(ZERO , RWRT)
WLV = WLVG + WLVD

RWLVG = GTOTAL * FLV - DLV
RWST = GTOTAL * FST
RWSO = GTOTAL * FSO
RWRT = GTOTAL * FRT

$$dW_{lv} / dt = RUE I_{int} F_{lv} - r_d W_{lv} \quad \text{Eq.(8)}$$

$$dW_{st} / dt = RUE I_{int} F_{st} \quad \text{Eq.(7)}$$

$$dW_{so} / dt = RUE I_{int} F_{so} \quad \text{Eq.(7)}$$

$$dW_{rt} / dt = RUE I_{int} F_{rt} \quad \text{Eq.(7)}$$



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Functions for spring wheat

TITLE LINTUL

FUNCTION FRTTB = 0.,0.50 , 110., 0.50 , ...
 275.,0.34 , 555.,0.12 , 780.,0.07 , ...
 1055.,0.03 , 1160.,0.02 , 1305.,0. , 2500.,0.

FUNCTION FLVTB = 0.,0.33 , 110.,0.33 , ...
 275.,0.46 , 555.,0.44 , 780.,0.14 , ...
 1055.,0. , 2500.,0.

FUNCTION FSTTB = 0.,0.17 , 110.,0.17 , ...
 275., 0.20 , 555., 0.44 , 780., 0.79 , ...
 1055., 0.97 , 1160., 0. , 2500., 0.

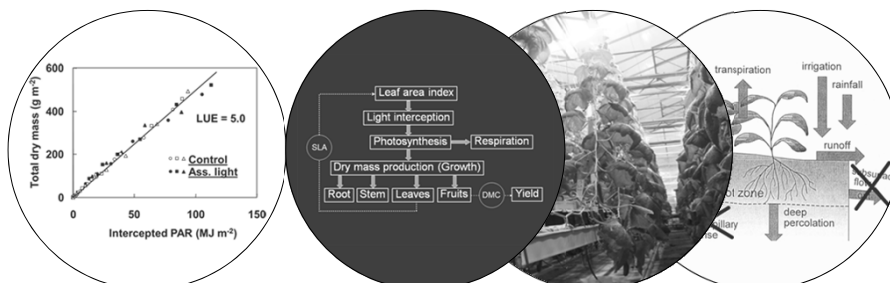
FUNCTION FSOTB = 0., 0. , ...
 1055.,0. , 1160.,0.98 , 1305.,1. , 2500.,1.



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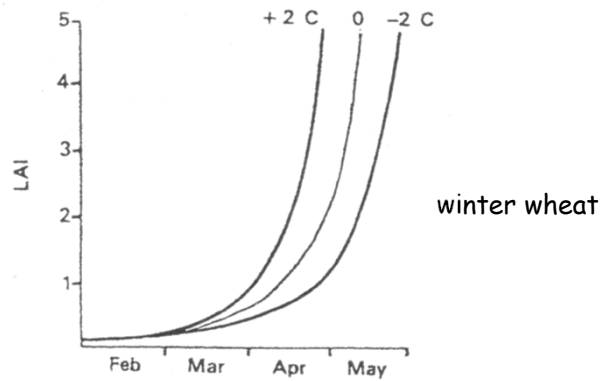
Lintul1: Potential crop growth

Early leaf growth



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Early leaf growth: temperature dependent



From: Monteith, 1966

Parameters for spring wheat & Growth of Leaf Area (GLA)

```
PARAM LAII = 0.012 ; SLA = 0.022 ; RGRL = 0.009 ; ...
      TSUMAN = 1110. ; LAICR = 4. ; RDRSHM = 0.03
FUNCTION RDRT = -10,0.03 , 10,0.03 , 15,0.04 , 30,0.09 , 50,0.09
```

```
*-----*
      SUBROUTINE GLA(TIME,DOYEM,DTEFF,TSUM,LAII,RGRL,DELT,
$      SLA ,LAI,GLV,
      IMPLICIT REAL (A-Z)
```

*---- Growth during **maturation** stage:

$$GLAI = SLA * GLV$$

$$dL/dt = S_{lo} F_{lv} RUE I_{int}$$

Eq.(7 & 13)

*---- Growth during **juvenile** stage:

IF ((TSUM.LT.330.) .AND. (LAI.LT.0.75))

$$GLAI = LAI * (EXP(RGRL * DTEFF * DELT) - 1.) / DELT$$

Eq.(12)

*---- Growth at day of seedling emergence:

IF ((TIME.GE.DOYEM) .AND. (LAI.EQ.0.))

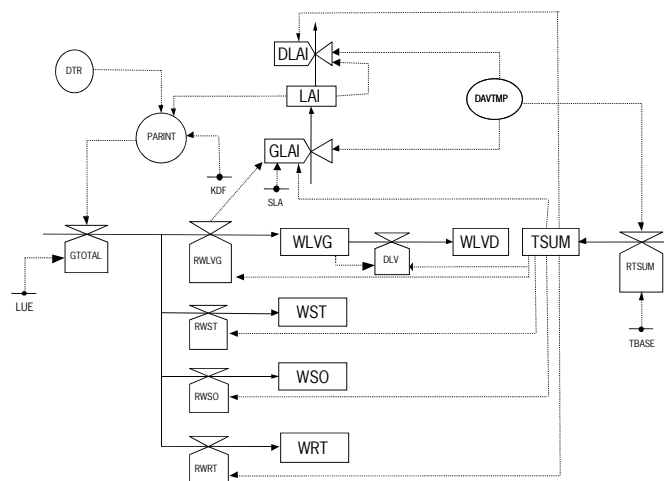
$$GLAI = LAII / DELT$$

*---- Growth before seedling emergence:

IF (TIME.LT.DOYEM) GLAI = 0.

```
RETURN
END
```

Summarizing relational diagram and meaning symbols



Forrester notation . . .

Symbol		Dimension
	Integral	Amount
	Differential equation (decision about rate)	Amount/time
	Flow of substance	Amount/time
	Discontinuous flow of substance	Amount/time
	Flow of information	No unit
	Input variable	Not known beforehand
	Auxiliary variable	Not known beforehand

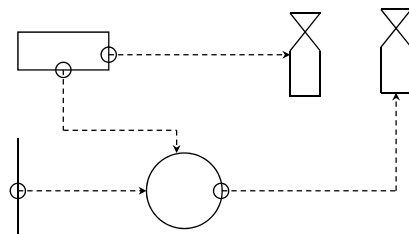


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From: Forrester, 1961

. . . Forrester notation (continued)

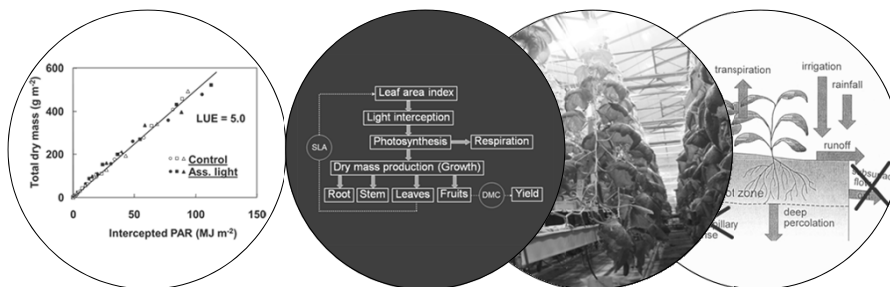
Information flows can only be used:
Information flows should point towards a
place of decision, never to a state:



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Lintul1: Potential crop growth

Exercises



Practical exercises Lintul1

Exercises on the following subjects:

- Consequence of weather and different emergence dates on crop growth
- The sensitivity of yield for some important crop parameters or, differently stated, the consequence of uncertainties in parameter values
- Effects of parameter changes in light extinction (light capture), and light use efficiency on growth
- Potential yield as a function of emergence dates

There are data provided for a dry, an average and a wet year

Choose between Brazil and the Netherlands

Prepare a report for yourself on the PC to prepare for the exam



To be able to do the exercises, study:

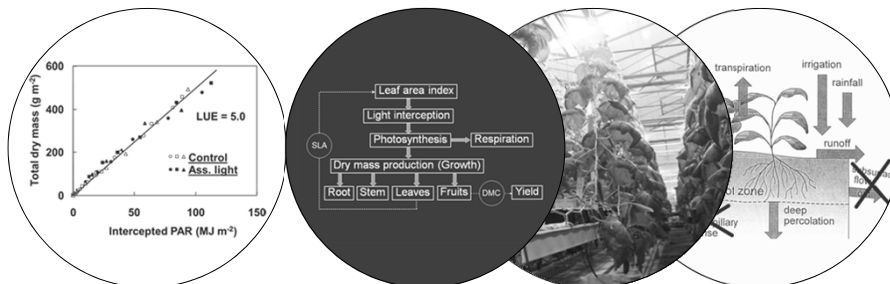
- Theory: "Lintul-1: potential crop growth"
- This power point presentation



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Lintul1: Potential crop growth

References



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